

Decoupling Safety Guarantees from Policy Competence in Multi-Robot Control Barrier Function Filters

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Abstract

Control barrier function (CBF) safety filters are increasingly used to wrap learned policies in multi-robot systems, on the premise that the filter makes the resulting system “safe.” We argue this premise conflates two distinct properties: velocity-level safety (the filter never commands a colliding action) and episode-level competence (the policy accomplishes the task). We formalize this distinction for a dual-lookahead CBF/ACAS filter applied to a differential-drive robot, and provide direct empirical evidence of the gap using a real multi-agent reinforcement learning (MARL) system: a centralized-training/decentralized-execution (CTDE) MAPPO policy for cooperative area coverage, trained under a short iteration budget that left it undertrained. Despite the safety filter being active on every control step and functioning as designed, the resulting policy still logs a higher-than-baseline environment-level collision rate (3.0 per episode vs. 0.8–2.4 for two non-learned baselines) and a mean swarm displacement of 0.0 m – i.e., the filter successfully keeps a stationary, incompetent policy from driving into obstacles, but cannot manufacture the exploratory behavior the policy never learned. We use this as a worked example to propose a simple three-class taxonomy for characterizing safety-filtered learned systems (competent-and-filtered, incompetent-and-filtered, unfiltered) and argue that reported “collision rate” figures for CBF-filtered policies should be accompanied by a competence metric, since the former alone cannot distinguish a genuinely safe capable system from a safe-by-paralysis one.

Keywords: control barrier functions, safe reinforcement learning, multi-robot systems, safety filtering, policy evaluation

1 Introduction

A control barrier function (CBF) renders a designated safe set forward invariant by solving, at each control step, a quadratic program (QP) that minimally perturbs a nominal control input subject to a barrier constraint [?]. In multi-robot and learning-enabled robotics, CBF filters are frequently proposed as a way to certify the safety of an otherwise uncertified learned policy: the policy proposes an action, the filter either passes it through or overrides it, and the composed system inherits a formal collision-avoidance guarantee independent of the policy’s internal quality [?]. Recent work continues to extend this pattern – enforcing CBF constraints *during* RL training rather than only at deployment [?], combining CBFs with actor-critic architectures via learned Koopman representations [?], and applying CBF-informed RL to connected and automated vehicles [?] and mobile robot navigation under risk [?].

This literature is, almost uniformly, framed around a positive claim: the filter makes the system safe. We do not dispute that claim at the level it is made – a correctly implemented CBF filter genuinely does bound the velocity-level action space to avoid detected collisions. Our contribution is narrower and, we argue, underexamined: we characterize what that guarantee does *not* imply, using a real, reproducible multi-agent system in which an undertrained policy remained “safe” under an active CBF filter while being functionally useless. We propose that this distinction – between velocity-level safety and episode-level competence – deserves explicit separation in how CBF-filtered learned systems are evaluated and reported, and offer a simple taxonomy for doing so.

2 Related Work

Ames et al. [?] establish the theoretical foundation for CBFs as a general safety-certification tool via forward invariance of a safe set. Cheng et al. [?] combine CBFs with RL by using the filter as an online safety layer around a learned policy for safety-critical continuous control, similar in spirit to the architecture we analyze; their emphasis, as in most follow-on work, is on demonstrating that the filter prevents constraint violation, not on characterizing the space of behaviors the filtered system can still exhibit when the underlying policy is weak. CBF-RL [?] internalizes CBF constraints into the training process itself so that safety need not be enforced online at deployment; this is a complementary direction to our analysis, which is specifically about systems that *do* retain an online filter, since online filtering remains common practice when training-time guarantees are unavailable or insufficient. Koopman-based CBF filters for actor-critic RL [?] and CBF-informed RL for connected and automated

vehicles [?] extend the filtering paradigm to new system classes and representations without revisiting the competence question. A risk-aware dynamic safety filter for mobile robot navigation [?] is closer to our concern in that it explicitly reasons about risk rather than binary safety, but does not report a competence metric alongside its safety metric. Yu et al. [?] establish MAPPO with a centralized critic as a strong CTDE baseline for cooperative multi-agent tasks; we use an architecture in this family as the learned policy under test, not as a contribution in itself.

3 System Under Test

We use a three-robot TurtleBot3 Waffle swarm performing cooperative area coverage in a Gazebo Sim environment (ROS 2 Jazzy), with a shared MAPPO policy trained via a centralized critic and a permutation-invariant neighbor-embedding actor. Full architectural detail is given in a companion system paper; here we summarize only what is needed to interpret the safety analysis.

Dual-lookahead CBF/ACAS filter. Given a nominal command $(v_{\text{nom}}, \omega_{\text{nom}})$ from the policy, the filter solves

$$\min_{v, \omega} (v - v_{\text{nom}})^2 + 0.05(\omega - \omega_{\text{nom}})^2 \text{ s.t. } \{A_k v + B_k \omega + \gamma h_k \geq 0\}_k, \quad (1)$$

over barrier constraints h_k derived from LiDAR returns and neighbor positions at two lookahead points $(\pm l, l = 0.12 \text{ m})$ along the robot’s heading axis, with safety margin $d_{\text{safe}} = 0.20 \text{ m}$ and class- $\mathcal{K}$ gain $\gamma = 2.0$, considering all LiDAR sectors and neighbors within 0.6 m . The QP is solved with SLSQP at every control step. On infeasibility, the filter halts forward motion while permitting rotation, rather than passing the unfiltered nominal command through. An auxiliary Active Collision Avoidance Safeguard (ACAS) commands a slow reverse whenever raw minimum LiDAR range drops below 0.22 m with positive commanded forward speed. This filter is applied identically regardless of which policy (learned or scripted) produced the nominal command, which is what makes it possible to isolate the filter’s contribution from the policy’s.

Environment-level collision metric. Following standard practice, we report a binary collision event whenever any robot’s minimum LiDAR range drops below 0.20 m during an episode – the same threshold used as d_{safe} in the filter. This is deliberately the metric most favorable to a “the filter makes it safe” narrative, since it is measured at the same distance the filter itself defends.

4 Empirical Characterization

We benchmark three methods under identical filter conditions in the nominal regime (5 episodes each): a Random Walk baseline, a reactive Frontier Heuristic baseline (closest-unvisited-cell target selection with reactive obstacle avoidance), and the MAPPO policy trained for 15 iterations (4,500 total environment steps – physics-in-the-loop training in Gazebo has no fast-simulator shortcut, so this is a genuinely small budget by MARL standards). All three methods are subject to the identical CBF/ACAS filter described in Section ??.

Table 1: Nominal-regime comparison, mean $\pm$ std over 5 episodes. All methods run under the identical CBF/ACAS filter.

Method	ACR (%) $\uparrow$	Redund. $\downarrow$	Dist. (m)	Coll. $\downarrow$
Random Walk	18.1 ± 0.6	52.64	6.3	0.80
Frontier Heuristic	21.1 ± 9.9	8.12	4.8	2.40
MAPPO (undertrained)	10.4 ± 1.5	14.63	0.0	3.00

The result central to this paper’s argument is the MAPPO row’s *mean displacement* of 0.0m co-occurring with its *highest* collision count of the three methods. Evaluated deterministically (no exploration noise), the policy collapses to a near-stationary behavior at or immediately after spawn – it does not explore the environment at all – while still logging more environment-level collision events than either scripted baseline, which do move. Diagnostic data from the training run corroborates the underlying cause: policy entropy fell to -0.53 by the final iteration (from a positive value earlier in training), and stochastic training rollouts had explored substantially more of the state space than the deterministic evaluation policy retained – consistent with a policy that converged to a locally “safe” near-stationary behavior before it had seen enough episodes to learn that exploration is rewarded.

Critically, this is not a filter failure. The filter is not designed to, and does not claim to, prevent the environment-level collision event that triggers when a stationary robot’s already-close neighbor or wall geometry places it under the 0.20m threshold at spawn, nor to invent an escape trajectory for a policy that never proposes one. The filter’s actual guarantee – bounded velocity given current sensing, with a rotation-only fallback on QP infeasibility rather than passing an unfiltered command – was exercised correctly throughout. What the result demonstrates is that this guarantee, though real, is insufficient on its own to produce the intuitive notion of a “safe robot”: a robot that is safe because it is frozen in place, adjacent to an obstacle it was placed near at spawn, is safe by the filter’s contract and useless by every other measure.

5 A Taxonomy for Reporting Filtered Systems

We propose classifying a CBF-filtered learned system along two independent axes rather than one: *filter status* (active/inactive, and whether it degrades gracefully on infeasibility) and *policy competence* (task-level performance under the filter, e.g. coverage rate, goal-reach rate, or an equivalent domain metric). This yields three practically distinguishable classes:

- **Class A – Competent and filtered.** The policy accomplishes the task and the filter is rarely or never binding. The intuitive “safe and capable” system.
- **Class B – Incompetent and filtered.** The policy fails to accomplish the task (low task-metric score, as in our 0.0m-displacement MAPPO run) while the filter prevents velocity-level constraint violation. Safe-by-paralysis or safe-by-restriction: the collision-rate metric alone cannot distinguish this from Class A.
- **Class C – Unfiltered.** No safety filter; task competence and safety are entangled, and any observed low collision rate is attributable to the policy alone.

Under this taxonomy, our MAPPO result is a documented, empirically produced Class B instance. We argue that any paper (including a system paper we have written using the same underlying platform) reporting a collision-rate figure for a CBF-filtered learned policy should also report a task-competence figure alongside it, precisely because Table ??’s collision column, read in isolation, does not distinguish MAPPO’s Class B result from a Class A one – the two are visually similar (low collision counts) but substantively opposite.

6 Discussion and Limitations

Our empirical evidence comes from a single system and a single undertrained policy; we do not claim the entropy-collapse mechanism that produced this specific Class B instance is the only route to one. We also do not claim CBF filters are ineffective – the filter in Section ?? performed exactly as specified throughout every experiment reported here, including the Class B case, and a policy-agnostic bounded-velocity guarantee is a genuinely useful property independent of the concerns raised in this paper. Our claim is narrower: that the guarantee and the outcome most readers associate with it (a capable, safe robot) are separable, that this separation is easy to produce by accident (an undertrained policy, not an adversarial one, was sufficient here), and that current reporting conventions in the CBF-filtered RL literature do not consistently surface it. A natural extension is a systematic sweep across training budgets, reward designs, and filter parameters to map the boundary between Class A and Class B behavior, rather than the single worked example provided here.

7 Conclusion

We characterized, with direct empirical evidence from a real multi-robot reinforcement learning system, the gap between what a CBF/ACAS safety filter guarantees (bounded, collision-avoiding velocity given current sensing) and what it is often implicitly credited with (a competent, safe robot). An undertrained MAPPO policy remained “safe” by the filter’s own contract while being entirely stationary and, by a strict environment-level metric, more collision-prone than two non-learned baselines that actually moved. We propose reporting task-competence alongside collision-rate for CBF-filtered learned systems, and offer a three-class taxonomy (competent-filtered, incompetent-filtered, unfiltered) as a starting point for making this distinction explicit in future work.

8 Postscript: Multi-World, Multi-Scale Safety Validation

The single-world, single-policy observation reported above motivated a broader validation of the underlying unicycle CBF/QP safety filter itself, independent of policy competence. The filter (separate obstacle and inter-agent safety radii, $d_{\text{safe,obs}} = 0.22$ m, $d_{\text{safe,agent}} = 0.45$ m, solved via OSQP at each control step) was evaluated across all five benchmark worlds (**cafe**, **warehouse**, **depot**, **office**, **maze**) under the Frontier Heuristic controller, at swarm sizes $N \in \{2, 3, 5, 8\}$, and separately under two dynamic (non-static) hazard scenarios.

Table 2: Inter-agent collision count across swarm-size sweep, 200-step episodes, 20 configurations.

World	$N=2$	$N=3$	$N=5$	$N=8$
cafe	0	0	2	8
warehouse	0	0	0	0
depot	0	0	0	0
office	0	0	0	2
maze	0	0	0	0

Inter-agent collisions remained at zero in 18 of 20 swarm-size configurations, with isolated contacts appearing only at $N=8$ in the two smallest arenas (**cafe**, 17×8 m; **office** corridors), consistent with the filter’s safety radius becoming a larger fraction of available free space as agent density increases, rather than with any breakdown of the filter’s guarantee. Under both dynamic-hazard tests – a head-on closing trajectory at $v_{\text{rel}} = 0.38$ m/s and an orthogonal crossing at $v_{\text{rel}} = 0.22$ m/s – the filter produced active braking and standstill-yielding behaviour respectively, holding minimum clearance above the configured

safety radius in both cases with zero contacts. Separately, root-cause analysis of a non-zero inter-agent collision count observed in the **depot** world (6 contacts, 3 consecutive control steps) traced the cause to sensor occlusion: two agents rounding opposite faces of the same convex structural pillar lost line-of-sight to each other until already inside the safety radius, after which the filter’s repulsion term separated them within the same control cycle. This is a sensing-visibility limitation of a purely reactive, single-step-lookahead filter, not a solver or constraint-encoding defect, and is a natural target for a longer-horizon or map-aware extension of the filter presented here.